

Benchmarking SLM and LLM in Feature Generation for Interpretable Automated Essay Scoring

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Abstract—In this paper, we explore feature generation with pretrained language models for automated essay scoring (AES). LLM have become increasingly popular in feature engineering, from feature selection through feature generation; however, in many domains, such as higher education, using proprietary LLM remains impossible because of data privacy restrictions or limited resources. We therefore benchmark the performance of several open-source small language models (SLM) as well as GPT-4o and Claude Sonnet 4.5 (as baselines) for feature generation on a commonly-studied AES dataset: the International English Language Testing System (IELTS). First, we have each model generate feature sets using a novel method for feature generation, inspired both by reasoning models and common practice within writing assessment. We then take a multifaceted approach to measuring the quality of the generated features, comparing the SLM and LLM features to traditional, statistical AES features and using the generated features to train several downstream models. Overall, we find that SLM are quite successful at feature selection and generation, and the most capable SLM were able to produce features nearly as useful for AES models as the tested LLM; however, the features are less correlated with traditional AES metrics than expected, raising questions over their faithfulness and reliability.

Index Terms—LLM, Small Language Models, Feature Engineering, Automated Essay Scoring

I. INTRODUCTION

Automated Essay Scoring (AES) often relies heavily on time-consuming, complex feature engineering, the process of deriving meaningful and predictive values from the text [1]. Traditional AES has used both surface-level characteristics, such as word or sentence counts, and more complex values, such as text embeddings, derived with deep learning.

While traditional feature engineering techniques have produced strong results in AES, the generated features generally remain either uninterpretable (such as embeddings created with BERT models) or only superficially related to essay quality. With manually engineered features, moreover, AES models also often remain tied to a specific essay prompt, unable to generalize across multiple writing assignments [2].

Recently, scholars have begun using LLM for various stages of the feature engineering process, from initial feature selection to feature generation. This work has shown that the vast

contextual knowledge possessed by LLM allow them to intuit which features might be useful in a classification problem [3]–[5] and even mine important features from text [6]. As many scholars have noted, these features, based in natural language, are interpretable, an especially important characteristic for AES [7]. While LLM have been explored often in AES [8]–[12], they have not yet been deployed for feature selection, and only recently have they been begun to show promise in trait-specific scoring, where these traits are then used as features in downstream, holistic scoring models [13], [14].

However, in the domain of education, the use of propriety, closed-source LLM, such as the OpenAI or Anthropic models, is often restricted, and privacy regulations preclude instructors and scholars from sharing student writing with these models. Open source, small language models (SLM) that can be run locally offer a potential solution, though SLM may not possess either the context knowledge or the deep reasoning ability to effectively generate features.

In this study, we benchmark the ability of several language models in feature engineering for AES, including both feature selection and feature generation. We first develop a novel method for generating interpretable features (or trait scores) from essays. We then evaluate the success of the features in multiple ways, including a comparison to traditional feature engineering approaches and the use of generated features in downstream AES models.

Our research is motivated by the following questions:

- Q1: Can language models generate features from student essays that can be used in downstream models for AES?
- Q2: How do features generated by SLM compare with those generated by LLM?
- Q3: Do the generated feature scores correlate with traditional metrics used in AES?

Overall, we find that even small models are quite adept at selecting meaningful features and can generate features useful in AES, though scores from models using these features lag behind state of the art AES approaches.

II. RELATED WORKS

A. Feature Engineering in Automated Essay Scoring

AES has a long history, dating back to the 1960s when Page developed the first algorithm to assess writing [15]. Since Page’s first system, AES research has largely focused on mining features from text that can be used in essay assessment, aiming at either holistic scoring [16] or, more recently, trait-based scoring [17]–[19]. [1] divide the features used into three categories: statistical features; style based features; and content based features. Initially, researchers relied on surface-level, statistical features, such as a count of words per sentence or a count of sentences per paragraph.

Deep learning approaches became increasingly common in the 2010s and then accelerated after the introduction of transformer models. Several researchers, for example, have incorporated BERT-based models into AES, using the embeddings generated by BERT as features or in the generation of novel features [20]–[22].

With the growing popularity of generative AI, much recent work has explored using LLM to grade essays. Scholarship has covered prompting strategies [16], [23]; feedback generation [24]; the use and effectiveness of rubrics [25]–[27]; and general reliability and benchmarking [8]–[12]. Early research found only a slight correlation between holistic scores generated by LLM and scores assigned by human graders, though more recent models have continued to improve.

While LLM have often been used to score essays, they have rarely been explored specifically for feature engineering within AES, a crucial step that can improve both interpretability and accuracy. However, they have been used for trait-based scoring, which can serve as a kind of feature generation if these traits are used in downstream holistic models. [13] recently demonstrated that trait-based scoring outperforms traditional, zero-shot holistic scoring, while [14] found success using rubric-based LLM-generated trait scores and combining these with traditional features in a downstream model.

B. Automated Feature Generation with LLM

Our research is novel in that we allow LLM and SLM to perform feature selection in addition to feature generation; that is, in previous studies, LLM generally mined text for holistic scores or predefined features, often assisted by rubrics and prompt engineering. In our approach, as we explain below, the language model first defines characteristics that might be predictive of a holistic score, develops a rubric for each characteristic, and scores a set of sample essays.

Recently, scholars have explored using LLM for feature engineering, capitalizing on the vast contextual knowledge acquired by the models during training. As [3] note, LLM have proven remarkably adept at predicting meaningful features when given the description of a prediction task. Several studies have shown promise for using LLM to determine or select these features [3]–[5].

Beyond feature selection, LLM have demonstrated the ability to generate features from text and perform traditional

feature engineering with tabular data [6] and multimodal data [28]. [7], for example, recently generated features from text using LLM across five diverse datasets. When used in downstream models, these features achieved performance similar to fine-tuned BERT models, with the added benefit, as the authors note, of feature interpretability. [29] similarly employed LLM to generate a set of human-interpretable features from datasets, which were again used to train models that outperformed various baselines. Other researchers have focused on enhancing the feature engineering capabilities of LLM, either by finetuning or prompting strategies [30].

Research has consistently shown a benefit from allowing LLM to oversee feature engineering, largely because the models handle context and domain knowledge skillfully [31]. We expect that the contextual knowledge gained during training should allow language models to effectively recognize meaningful characteristics or traits in an essay and derive those features from text.

III. METHODOLOGY

A. Automated Feature Generation

In this study, we used the IELTS foreign language exam essays, a dataset from a recent Kaggle competition for AES.¹ This dataset includes about 1,200 short essays which have been scored on a scale of 1 to 10.

The test is evaluated for both an overall score and four characteristics (Task Achievement, Coherence and Cohesion, Lexical Resource, and Grammatical Range and Accuracy). Our initial dataset, however, contained only holistic scores.

To generate features from the essays, we employed the three step process below, which draws on both chain-of-thought prompting and traditional practice within writing assessment. As in [27], this can be seen as a multi-agent system, with separate prompts for each stage in the process.

- 1) First, we prompted the model to specify a list of n features useful for assessing IELTS essays. In the prompt, we included information about the model’s role and the IELTS foreign language exam essay. We also provided one of the top-score essays as a good example and one of the low-score essays as a bad example. Examples were included to provide additional context for the model, which, we reasoned, would likely improve both its ability to extract relevant features and its ability to generalize across prompts.
- 2) Next, we looped through the list of suggested features and had the model develop a rubric with five categories for each feature: poor, below average, average, good, and excellent. We used these descriptive grades rather than numeric grades because we found SLM inclined to grant very high scores when assigning numeric grades.
- 3) Finally, we fed the model the feature name and feature rubric and had it evaluate sample essays. We looped through the list of features and generated a score for each feature.

¹See <https://www.kaggle.com/code/yunsuxiaozi/ielts-writing-scored-essays>

We ran this process for (n=) four to eight features, since feature suggestions became redundant after n=8, and we tested the following models: SMOL-1.7B Instruct, Gemma 3 4b, Phi-4-mini-instruct, Qwen 3 4b, Qwen 3 1.7B, Qwen 3 600m, Llama 3.2 3b-Instruct, GPT-4o, and Claude Sonnet 4.5.

B. Measuring Feature Quality

We then measured the quality of features produced through a multifaceted approach.

First, we compared the generated feature scores to scores used commonly in AES, derived both through surface-level statistics and traditional NLP. For this, we recorded the following features for each essay:

- Flesch Kincaid Reading Level, which is calculated by the following formula: $(0.39 \times \text{Avg Sentence Length}) + (11.8 \times \text{Avg Syllables per Word}) - 15.59$.
- Vocabulary complexity, which we measured as simply the number of unique words divided by the number of total words in an essay.
- A count of spelling errors.
- The average length (number of words) of each sentence.
- The number of complex sentences.
- A count of grammatical errors.
- The Gunning Fog index which is calculated $0.4 \times (\text{Average Sentence Length} + \text{Percentage of Complex Words})$
- The Guiraud Index, a measure of vocabulary complexity, which is calculated as the number of unique words over the square root of the number of total words in an essay.
- Flesch Reading Ease, which uses the formula $206.835 - (1.015 \times \text{Average Sentence Length}) - (84.6 \times \text{Average Syllables per Word})$
- The ARI Index, calculated as $4.71 \times (\text{characters/words}) + 0.5 \times (\text{words/sentences}) - 21.43$
- A count of adposition words, derived using the Spacy package.
- A count of coordinating conjunctions, derived using the Spacy package.
- A count of nouns, derived using the Spacy package
- A count of auxiliary verbs, derived using the Spacy package
- A count of verbs, derived using the Spacy package
- A count of adjectives, derived using the Spacy package
- A count of adverbs, derived using the Spacy package
- A count of pronouns, derived using the Spacy package

We then analyzed the correlation between each of these traditional features and the SLM or LLM generated features. For example, many language models created a feature related to vocabulary ("Academic Vocabulary", "Vocabulary Usage and Accuracy"), which we compared to the traditional measure for vocabulary complexity.

Next, we explored whether we could train regression models successfully on the generated features. We divided the set of 1,200 scored essays into a train set of 1,000 essays and a test set of 200. We then trained three regression models: Linear Regression, XGBoost, and LightGBM. For each, we recorded the MAPE, MAE, MSE, Pearson Correlation, and Quadratic

Weighted Kappa scores. The last two are used most frequently in AES, measuring the correlation between human scores and model scores, and are defined as follow:

The Pearson Correlation quantifies the linear relationship between two variables. It ranges from 1 (perfect correlation) to -1 (perfect negative correlation), with 0 indicating no correlation. It is defined as the covariance between x and y over the product of the standard deviation of x and the standard deviation of y.

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Quadratic Weighted Kappa (QWK) measures agreement between two raters in ordinal categories, taking into account not simply whether the ratings agree or disagree, but how far apart the ratings are. To calculate QWK, we create a confusion matrix showing how many times each rating pair occurred ($O_{i,j}$), an expected matrix with the number agreements that would happen randomly by chance ($E_{i,j}$), and a weight matrix that assigns a penalty by the distance between raters ($W_{i,j}$). QWK is then defined as follows:

$$\kappa = 1 - \frac{\sum_{i,j} W_{i,j} O_{i,j}}{\sum_{i,j} W_{i,j} E_{i,j}}$$

As a baseline, we also trained a vanilla BERT model on the train set, scored the test set, and recorded the same metrics.

We then determine feature importance for each model using two methods. For the LightGBM and XGBoost models, we apply TreeSHAP to derive the global features scores and determine the most significant generated features; for Linear Regression, we simply recorded coefficients.

Finally, we combined all features produced by SLM (from the n=8 run) into a single dataset. We used forward and backward selection for each model to determine which features to include and trained models on each selected feature set, once again testing the process with Linear Regression, XGBoost, and LightGBM and optimizing for R-squared. We then added the traditional, statistical features to the all-features dataset and repeated the process of forward and backward selection.

IV. RESULTS

A. Features and Rubrics

All models were able to describe useful features and create rubrics related to the features. However, only GPT-4o selected features that matched the criteria used in the IELTS, suggesting that its training context allowed it to tailor feature selection directly to the task. Feature categories were, on the whole, consistent across both LLM and SLM, though specific features varied. Table I below shows all unique features generated by each model.

Overall, both SLM and LLM also generated high quality rubrics, though SLM rubrics were occasionally plagued by some common failings, including repetition, lack of clear examples, and inaccuracies or hallucinations.

TABLE I
UNIQUE FEATURES GENERATED ACROSS CONFIGURATIONS BY EACH LANGUAGE MODEL

Model	Unique Features Generated
claude-sonnet-4.5	Argument Coherence, Argument Structure, Cohesion and Coherence, Evidence Quality, Grammatical Accuracy, Grammatical Precision, Lexical Range, Paragraph Development, Position Clarity, Sentence Fluency, Structural Organization, Task Response, Task Response Relevance, Thesis Clarity, Vocabulary Sophistication.
gemma 3-4b-it	Argumentation Quality, Argumentation Strength, Coherence and Cohesion, Collegiate Writing, Grammatical Accuracy, Lexical Diversity, Originality, Sentence Complexity, Sentence Structure Complexity, Topical Development.
gpt-4o	Argument Development and Support, Coherence and Cohesion, Conclusion Effectiveness, Critical Thinking and Originality, Formal Tone and Style, Grammatical Range and Accuracy, Language Proficiency and Vocabulary, Lexical Resource, Originality and Insight, Task Achievement, Task Response.
Llama 3.2-38-Instruct	Adherence to Task, Argument Development, Coherence, Collegiate Writing, Originality, Sentence Structure, Sentence Structure Diversity, Supporting Evidence, Tone and Tone Shifts, Tone and Voice, Use of Transitions and Connectors, Vocabulary Range.
Phi-4-mini-instruct	Argument Strength, Argument Strength and Clarity, Coherence and Cohesion, Engagement and Interest, Engagement and Style, Grammar and Mechanics, Grammar and Punctuation Accuracy, Originality, Originality and Creativity, Relevance, Relevance and Focus, Structure and Organization, Vocabulary Range, Vocabulary Range and Accuracy.
Qwen-0.6B	Credibility of Sources, Cultural Relevance, Depth of Analysis, Grammar and Syntax, Grammar and Syntax Accuracy, Innovation in Topic, Originality, Originality Collegiate Writing, Relevance to the Essay's Context, Structure and Organization, Use of Academic Vocabulary, Use of Complex Language, Use of Evidence, Use of Technical Terminology, Use of Writing Techniques.
Qwen3-1.7B	Academic Tone and Style, Clarity and Conciseness, Clarity and Precision, Clarity of Expression, Coherence and Logical Flow, Coherence and Structure, Grammar and Syntax, Originality, Originality and Innovation, Thesis Statement and Conclusion, Thesis Statement and Focus, Use of Academic Language, Use of Examples and Evidence, Word Choice and Diction, Word Choice and Precision.
Qwen3-4B	Coherence, Coherence and Cohesion, Cultural Relevance, Grammar Accuracy, Grammatical Accuracy, Lexical Resource, Logical Flow, Originality, Originality and Depth, Overall Structure, Sentence Variety, Topic Control, Topic Relevance, Vocabulary Range.
SMOL-1.7B	Academic Vocabulary, Cohesion, Cohesion and Coherence, Critical Thinking, Critical Thinking and Analysis, Engagement and Style, Grammar and Mechanics, Originality, Sentence Structure, Sentence Structure Variety, Tone and Voice, Use of Active Voice, Use of Rhetorical Devices, Use of Transitions.

B. Feature Correlation

When exploring the correlation between GenAI features and traditional features, we found less overlap than expected. Even the LLM produced features that showed only weak correlation to traditional metrics.

We can divide related features into three primary groups: those focused on vocabulary (Table II); those focused on grammar and spelling (Table III); and those focused on complexity (Table IV). Overall, we did detect a correlation between counts of grammar errors and related generated metrics across multiple models, though this score was below .5 for all models except GPT-4o. The smallest models (anything under 2B parameters) showed almost no correlation between generated scores for grammatical accuracy and a count of grammar errors. Vocabulary complexity and sentence complexity showed little to no correlation across all models.

TABLE II
FEATURES V. TRADITIONAL FEATURES: VOCABULARY

Model	Feature	ADJ	ADV	Vocab
claude-sonnet-4.5	Vocabulary Sophistication	0.18	0.14	0.15
gemma-3-4b-it	Lexical Diversity	0.15	0.13	0.09
gpt-4o	Lexical Resource Language Proficiency and Vocabulary	0.10 0.08	0.09 0.07	0.13 0.09
Llama-3.2-3B-Instruct	Vocabulary Range	0.37	0.35	0.07
Phi-4-mini-instruct	Vocabulary Range Vocabulary Range and Accuracy	0.59 0.42	0.49 0.30	0.21 0.10
Qwen3-0.6B	Use of Academic Vocabulary Use of Complex Language Use of Complex Vocabulary	0.19 0.20 0.41	0.19 0.19 0.35	-0.02 -0.05 0.00
Qwen3-1.7B	Use of Academic Language Word Choice and Diction Word Choice and Precision	0.30 0.36 0.29	0.21 0.30 0.27	0.18 0.16 0.07
Qwen3-4B	Lexical Resource Vocabulary Range	0.24 0.40	0.20 0.36	0.11 0.16
SMOL-1.7B	Academic Vocabulary	0.29	0.25	0.03

C. Model Quality

While models trained on the features generated by LLM were most successful, some of the small models, especially those around four billion parameters, were able to produce meaningful features that could be used to train highly accurate models. Qwen 3 4B was the most successful SLM, with metrics lagging only slightly behind GPT-4o and Sonnet-4.5. The results for all models trained can be found in Table V.

Even our baseline BERT model produced strong results, with an MSE of .66, a Pearson's Correlation of .54, and a

TABLE III
MODEL FEATURES V. TRADITIONAL FEATURES: GRAMMAR

Model	Feature	Grammar Errors	Spelling Errors
claude-sonnet-4-5	Grammatical Accuracy	-0.48	0.04
	Grammatical Precision	-0.50	0.02
gemma-3-4b-it	Grammatical Accuracy	-0.22	0.14
	Grammatical Accuracy	-0.22	0.14
gpt-4o	Grammatical Range and Accuracy	-0.56	-0.01
Phi-4-mini-instruct	Grammar and Punctuation Accuracy	-0.21	0.15
Qwen3-0.6B	Grammar and Syntax	0.03	-0.06
	Grammar and Syntax Accuracy	-0.08	-0.15
Qwen3-1.7B	Grammar and Syntax	-0.05	0.12
Qwen3-4B	Grammar Accuracy	-0.34	0.14
	Grammatical Accuracy	-0.31	0.17
SMOL-1.7B	Grammar and Mechanics	-0.02	0.21

QWK of .69. As Table V (which records the scores for the top feature set for each model) shows, multiple models improved significantly on the MSE, though the Pearson Correlation and QWK could not match the vanilla BERT baseline.

Finally, our models produced by forward and backward selection found between 13 and 49 of the features generated by SLM useful. The scores for these models were higher than the scores from models produced by any individual SLM. Although some traditional features improved this performance, SLM-generated features were among the most important, according to both the coefficients and the Shapley values. In the XGBoost model trained with backward selection (the model achieving the best MSE), Qwen 3 4B produced by far the most important feature (Grammar Accuracy), and two other in the top ten (Coherence, Sentence Variety), while Llama 3.2 3B contributed another top ten feature (Sentence Structure). Extending to the top twenty features, according to Shapley Values, Qwen 3 0.6B, Phi 4-mini-instruct, and SMOL 1.7B all contributed features.

V. DISCUSSION

Features selected and then generated by both LLM and SLM were remarkably consistent and germane to the task of essay assessment. Despite some difference in vocabulary, almost all models prioritized traditional, surface-level metrics such as

vocabulary and grammar, as well as more complex metrics like cohesion, critical thinking, and originality. As models were asked to name additional features, however, they became less relevant and more abstract and repetitive. SMOL-1.7B, for example, included both "Tone and Voice" and "Use of Active Voice" in its list of eight features.

We obtained mixed results for the quality and usefulness of the generated features, not only from the SLM but also the LLM. First, most of our features were useful in training essay scoring models. Almost all the trained models had an MSE near or lower than the baseline, though the QWK and Pearson Correlation lagged behind the BERT model. In part, this resulted from highly imbalanced scores in the dataset, with 53% of the essays scoring either a 6 or 7. Our trained models responded to this, resulting in low MSE, though this meant that less of the grade variation was explained by the model.

Smaller models, especially, were driven by simple features, particularly grammatical accuracy, though grammar was important for the scores of models trained with LLM-generated features, as well. More abstract features, like "Originality," were less important in both the models trained with LLM-generated and SLM-generated features.

Despite the limited gains, our tests with forward and backward selection found several of the features generated by SLM useful for the prediction task, and a mixed model approach outperformed any of the individual SLM. This "panel of experts" produced the best overall MSE (0.51, when using Backward selection and XGBoost), and achieved QWK and Pearson Correlation scores nearly matching scores of the models using LLM-generated features. In our best panel of experts model, six different SLM contributed to the model, though only Qwen 3 4B had a feature more significant than traditional, statistical features. This suggests the potential for blending smaller models in AES, which would allow instructors and researchers to keep the process local and confidential while achieving near-LLM results.

In contrast, our generated features showed only weak correlation with traditional features, raising questions over the faithfulness and interpretability of the results. Admittedly, this could indicate limitations with traditional, surface-level features; for example, a short essay may use a number of different small words, thus having a high percentage of unique words to total words while still showing a limited vocabulary. If this were the case, in longer essays, we would expect the correlation to increase, but essays written for the IELTS test tend to be very short, only around 200-300 words.

VI. CONCLUSION

In this paper, we have demonstrated that even SLM thrive in feature selection for AES, and they can use context knowledge to select and generate meaningful features for scoring IELTS essays. As noted above, assigning feature selection to a model as the first step in an AES pipeline will make for a more robust, task-independent scorer and draws on training knowledge possessed by these models. It would also allow for more

TABLE IV
MODEL SENTENCE COMPLEXITY METRICS

Model	Metric Detail	Sent Len	CCONJ	Complex Sent	Flesh RE	Flesh-K	Guiraud Index	Gunning Fog
gemma-3-4b-it	Sentence Complexity	-0.09	0.16	0.36	0.43	0.16	0.43	0.06
	Sentence Structure Complexity	-0.12	0.19	0.41	0.47	0.15	0.47	0.05
Llama-3.2-3B-Instruct	Collegiate Writing	-0.06	0.20	0.42	0.42	0.17	0.42	0.09
	Use of Transitions and Connectors	-0.07	0.09	0.25	0.24	0.11	0.24	0.05
Qwen3-0.6B	Collegiate Writing	0.03	0.14	0.20	0.22	0.09	0.22	0.06
Qwen3-4B	Overall Structure	-0.12	0.19	0.41	0.45	0.10	0.45	0.01
	Sentence Variety	-0.09	0.21	0.40	0.46	0.13	0.46	0.05
SMOL-1.7B	Sentence Structure and Variety	0.05	0.21	0.26	0.28	0.16	0.28	0.11
	"Sentence Structure"	0.00	0.14	0.23	0.22	0.11	0.22	0.05

TABLE V
MODEL EVALUATION METRICS

	Logistic Regression			Light GBM			XGBoost		
	Mean Squared Error	Pearson Correlation	Weighted Quadratic Kappa	Mean Squared Error	Pearson Correlation	Weighted Quadratic Kappa	Mean Squared Error	Pearson Correlation	Weighted Quadratic Kappa
claude-sonnet-4-5									
7	0.57	0.63	0.54	0.58	0.62	0.51	0.58	0.62	0.50
gemma 3-4b									
8	0.76	0.44	0.32	0.8	0.38	0.25	0.8	0.44	0.36
GPT-4o									
6	0.56	0.64	0.53	0.59	0.61	0.52	0.54	0.66	0.56
Llama-3.2 3B-Instruct									
8	0.81	0.38	0.24	0.82	0.36	0.23	0.82	0.4	0.28
Phi-4-mini-instruct									
8	0.74	0.46	0.31	0.8	0.38	0.21	0.74	0.48	0.4
Qwen 3-600M									
7	0.86	0.29	0.17	0.92	0.12	0.07	0.9	0.37	0.28
Qwen 3 - 1.7B									
7	0.73	0.47	0.37	0.8	0.39	0.27	0.72	0.51	0.44
Qwen 3-4B									
8	0.65	0.55	0.46	0.69	0.52	0.4	0.65	0.58	0.51
SMOL-1.7B									
8	0.85	0.32	0.18	0.93	0.13	0.07	0.86	0.38	0.31
Forward/Backward Selection									
Forward - SLM	0.61	0.49	0.38	0.70	0.48	0.40	0.60	0.54	0.49
Backward - SLM	0.61	0.49	0.39	0.66	0.51	0.46	0.64	0.51	0.43
Forward - SLM & Trad	0.58	0.53	0.43	0.62	0.52	0.43	0.53	0.59	0.50
Backward - SLM & Trad	0.58	0.52	0.39	0.57	0.58	0.48	0.51	0.61	0.50

Note: For comparison, a fine-tuned BERT baseline achieves: MSE = 0.66, Pearson = 0.54, and QWK = 0.69.

transparent and interpretable scoring, as we are able to see which of the generated features contribute most to any score.

We also saw that some SLM, especially Qwen 3 4B, generated features nearly as useful for AES as LLM such as GPT 4o and Claude Sonnet 4.5. Moreover, by combining the features of multiple SLM, we were able to produce models that matched models trained on LLM generated features, offering the possibility of using local, secure and efficient language models in an AES task.

However, when comparing the generated features to traditional statistical features, we saw less correlation than ex-

pected; while the features proved meaningful, the models struggled to generate simple features, like grammatical accuracy or vocabulary complexity, that correlated with traditional metrics for measuring these characteristics in writing. At the very least, this calls for additional research into the faithfulness of these generated features; while [7] and [29] have noted the potential of LLM-generated features for rendering models more interpretable, our research points to questions over the reliability and thus interpretability of these generated features.

SLM and LLM possess great potential for AES, but further work is needed to ensure our AES algorithms are fair, trans-

parent, secure, and responsive to various writing tasks and contexts. In future research, we hope to further investigate both the faithfulness and interpretability of SLM generated features and the ability of SLM to generate feedback along with scores.

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